

MBD4 loss defines a Triaxis therapeutic model spanning cytidine analogs, checkpoint inhibition, and immune checkpoint blockade

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Abstract

MBD4 loss-of-function (LOF) creates three therapeutically distinct consequences from a single base-excision-repair defect. Axis 1 is causal cytidine-analog synthetic lethality: gemcitabine or cytarabine increases unrepaired substrate and stalls replication forks. Axis 2 is replication-stress checkpoint dependence: MBD4-LOF models show ceralasertib sensitivity across a pan-cancer cohort, with the strongest response in TP53-mutant disease, and a class-concordant adavosertib signal. Axis 3 is hypermutator-directed immune checkpoint blockade: unrepaired methyl-CpG deamination fixes as CpG>TpG transitions, creating an outlier mutational state linked to anti-PD1 response in metastatic uveal melanoma.

The PARP branch is separate. MBD4-LOF does not increase baseline PARP1 expression (median 6.77 vs 6.66 log1p TPM; $p=0.605$; $n=19$ vs 1,498) and does not meaningfully separate PARP-inhibitor response (Δ median $Z=-0.024$). CrisPRO therefore treats the 7.41 log1p TPM Q75 value as a dataset-derived exploratory companion-diagnostic boundary: sub-threshold or missing expression triggers `HARD_BLOCK_LACKS_TRAPPING_SUBSTRATE` within the trial-prioritization engine, while independently validated BRCA/HRD evidence preserves PARPi routing.

The canonical GDSC2 ceralasertib comparison shows a 2.08-fold lower geometric-mean IC50 in MBD4-LOF models ($n=14$ vs 914; $p=0.0215$; $d=-0.503$). MSS restriction strengthens the signal ($n=10$ vs 906; $p=0.0153$; $d=-0.623$), and TP53-mutant stratification reaches a 2.91-fold reduction ($n=11$ vs 619; $p=0.0030$; $d=-0.740$). In the Saint-Ghislain metastatic uveal melanoma cohort, 3 of 5 MBD4-mutated tumors responded to immune checkpoint inhibition versus 4% of MBD4-wild-type tumors ($p<0.001$); median progression-free survival was 22.3 versus 4.0 months ($HR=0.22$; $p=0.01$), and median overall survival was unreached versus 16.6 months ($HR=0.11$; 95% CI 0.02–0.86; $p=0.04$). The Triaxis

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model encodes evidence-tiered actions for cytidine analogs, ATR/WEE1 trials, and hypermutator-confirmed immune checkpoint blockade.

Introduction

MBD4 repairs thymine and uracil mismatches generated by spontaneous deamination at methylated CpG sites (Hendrich et al. 1999). Its loss leaves two linked molecular products: unrepaired base lesions that obstruct DNA replication and CpG>TpG transition mutations that accumulate across the tumor genome (Bader et al. 1999; Krokan and Bjoras 2013; Wallace et al. 2012). Those products define three therapeutic routes rather than a single generic DNA-repair phenotype.

Axis 1 exploits the lesion substrate directly. Isogenic MBD4 knockout produces gemcitabine and cytarabine hypersensitivity, MBD4 re-expression restores resistance, and in vivo models preserve the effect (Chabot et al. 2022). Axis 2 exploits the checkpoint machinery required to stabilize lesion-stalled forks. ATR/CHK1/WEE1 signaling becomes the survival circuit for cells carrying unresolved BER damage, creating a pharmacologic route to ceralasertib and a class-concordant WEE1-inhibitor expansion. Axis 3 exploits the mutational consequence of failed methyl-CpG repair. CpG>TpG hypermutation increases the repertoire of tumor-specific altered peptides that can be presented as neoantigens, providing a mechanistic bridge to immune checkpoint blockade.

Human evidence distinguishes Axis 3 from the cell-line pharmacology supporting Axis 2. Rodrigues et al. reported an exceptional anti-PD1 response in metastatic uveal melanoma with germline deleterious MBD4 alteration, somatic MBD4 inactivation, and an outlier CpG>TpG mutation burden (Rodrigues et al. 2018). Saint-Ghislain et al. then evaluated MBD4 in a retrospective immune-checkpoint-treated metastatic uveal melanoma cohort and reported higher objective response, progression-free survival, and overall survival among MBD4-mutated tumors (Saint-Ghislain et al. 2022). This evidence establishes a clinical immunotherapy pillar while retaining its tissue and sample-size boundaries.

The PARP model requires separate adjudication. MBD4 loss would need to create a PARP1-high trapping state to function as a standalone PARPi biomarker. DepMap expression and GDSC2 response data test that requirement directly. The resulting CrisPRO Triaxis architecture therefore combines three positive evidence routes—cytidine analog synthetic lethality, replication-stress checkpoint inhibition, and hypermutator-directed immune checkpoint blockade—with an expression-gated safeguard against unsupported MBD4-only PARPi assignment.

Results

Axis 1: Cytidine analog synthetic lethality

In isogenic HAP1 models, MBD4 knockout increases gemcitabine sensitivity approximately 10-fold, reducing IC50 from 20.1 nM to 2.3 nM ($p=2.82\times 10^{-3}$). Re-expression restores resistance, and PDX evidence carries the response in vivo (Chabot et al. 2022). These experiments identify MBD4 loss as the causal determinant of cytidine-analog sensitivity and establish CYTIDINE_ANALOG_SYNTHETIC_LETHALITY as the Axis 1 action.

PARP1 expression does not connect MBD4 loss to PARP-inhibitor response

PARP1 expression is unchanged in MBD4 True-LOF lines relative to the non-LOF pool (median 6.77 vs 6.66 log1p TPM; $p=0.605$; $n=19$ vs 1,498). RNF144A is also unchanged (median 2.15 vs 1.71; $p=0.476$). Direct pharmacologic stratification produces no meaningful MBD4-specific PARPi response separation (Δ median $Z=-0.024$; bootstrap 95% CI -0.838 to $+0.624$; $n=8$ LOF). In contrast, baseline PARP1 expression predicts PARPi response across 481 matched lines (Spearman $\rho=-0.416$; $p=1.36\times 10^{-21}$).

CrisPRO converts this decoupling into an exploratory companion-diagnostic framework. The DepMap-derived Q75 boundary of 7.41 log1p TPM is not a validated clinical cutoff. Within the trial-prioritization engine, missing or sub-Q75 PARP1 executes HARD_BLOCK_LACKS_TRAPPING_SUBSTRATE, expression at or above Q75 executes ALLOW_PARPI_TRIAL_EVALUATION, and pathogenic BRCA1/2 or another validated independent indication executes ALLOW_PARPI_ROUTING_BYPASS.

Axis 2: Replication-stress checkpoint inhibition

MBD4-LikelyLoF spans Bowel, Lymphoid, Ovary, Uterus, Esophagus, CNS, Prostate, and Lung. In the canonical GDSC2 comparison, ceralasertib shifts LN_IC50 by -0.732 ($p=0.0215$; $d=-0.503$; $n=14$ MBD4 True-LOF vs 914 WT), corresponding to a 2.08-fold lower geometric-mean IC50. Removing MSI-H models strengthens the effect to Δ LN_IC50 $=-0.910$ ($p=0.0153$; $d=-0.623$; $n=10$ vs 906). Every leave-one-out iteration remains below $p=0.05$. The all-lineage route maps to HIGH_CONFIDENCE_TRIAL_CANDIDATE; the MSS route maps to PRIORITIZE_ATRI_ENROLLMENT.

Adavosertib extends the response across the ATR/CHK1/WEE1 cascade (Δ LN_IC50 $=-0.512$; $p=0.0733$; $d=-0.361$; $n=15$ vs 929), a 1.67-fold lower geometric-mean IC50. CrisPRO maps this class-concordant secondary to CLASS_CONCORDANT_WEE1I_SECONDARY.

TP53-mutant enrichment Within TP53-mutant models, MBD4-LOF produces the largest ceralasertib effect: Δ LN_IC50 $=-1.069$ ($p=0.0030$; $d=-0.740$;

n=11 vs 619), equivalent to a 2.91-fold lower geometric-mean IC50. This state maps to `HIGH_CONFIDENCE_TRIAL_ENROLLMENT` and defines the primary HG-SOC enrichment route because TP53 mutation is present in approximately 96% of HGSOC.

Colorectal expansion The canonical bowel comparison contains n=5 MBD4-LOF and n=41 WT models. Ceralasertib produces Δ LN_IC50=-0.692 (p=0.126; d=-0.464), a 2.00-fold lower geometric-mean IC50. The effect-size-defined prospective expansion maps to `COLORECTAL_TRIAL_TARGET`.

Axis 3: CpG>TpG hypermutator phenotype and clinical immune checkpoint blockade

MBD4 loss prevents excision of thymine generated by 5-methylcytosine deamination. Replication across the unrepaired G:T mismatch fixes a C>T substitution, producing the CpG>TpG transition signature and an outlier tumor mutation burden. The expanded altered-peptide repertoire supplies candidate neoantigens for immune recognition; checkpoint blockade can release T-cell responses against that immunogenic substrate. This mechanism links MBD4 enzymatic failure to immunotherapy without treating mutation burden alone as proof of response.

Rodrigues et al. reported the sentinel human case: metastatic uveal melanoma with a deleterious germline MBD4 mutation, somatic inactivation of the remaining allele, outlier CpG>TpG mutation burden, and exceptional sensitivity to anti-PD1 therapy (Rodrigues et al. 2018). Saint-Ghislain et al. provided retrospective cohort evidence (Saint-Ghislain et al. 2022). Among 300 immune-checkpoint-treated metastatic uveal melanoma patients, the all-comer objective response rate was 3.3%. MBD4 was sequenced in 131 tumors; five (3.8%) were mutated. Three of five MBD4-mutated tumors responded versus 4% of MBD4-wild-type tumors (60% vs 4%; p<0.001). Median progression-free survival was 22.3 months in MBD4-mutated disease versus 4.0 months in wild type (HR=0.22; p=0.01). Median overall survival was unreached versus 16.6 months (HR=0.11; 95% CI 0.02–0.86; p=0.04).

These patient outcomes establish Axis 3 as the human-clinical pillar of the Triaxis model. Its action, `IMMUNOTHERAPY_CHECKPOINT_BLOCKADE`, requires an MBD4-LikelyLoF state plus a confirmed CpG>TpG hypermutator signature or validated equivalent; MBD4 status alone does not substitute for tumor mutational profiling. The current cohort is retrospective and contains five MBD4-mutated tumors, so prospective tissue-agnostic validation remains part of the action definition rather than an omitted boundary.

Heterozygous LikelyLoF as an operational genomic gate

Twenty of 21 default MBD4-LikelyLoF models (95.2%) carry heterozygous genotype calls; one is homozygous-alt. This composition supports inclusion of single-

allele LikelyLoF states in prospective trials without requiring completed LOH testing before enrollment. The drug-matched cohort does not resolve allele-specific function, so the gate does not prove haploinsufficiency.

Triaxis clinical routing matrix

Biomarker state	Locked evidence	CrisPRO action
MBD4 LikelyLoF	Isogenic knockout/rescue and PDX cytidine-analog axis	CYTIDINE_ANALOG_SYNTHETIC_LETHALITY
MBD4-LOF, all lineages	Ceralasertib n=14 vs 914; $\Delta=-0.732$; p=0.0215; d=-0.503	HIGH_CONFIDENCE_TRIAL_CANDIDATE
MBD4-LOF, MSS	Ceralasertib n=10 vs 906; $\Delta=-0.910$; p=0.0153; d=-0.623	PRIORITIZE_ATRI_ENROLLMENT
MBD4-LOF/TP53-mutant	Ceralasertib n=11 vs 619; $\Delta=-1.069$; p=0.0030; d=-0.740	HIGH_CONFIDENCE_TRIAL_ENROLLMENT
MBD4-LOF colorectal/bowel	Ceralasertib n=5 vs 41; $\Delta=-0.692$; p=0.126; d=-0.464	COLORECTAL_TRIAL_TARGET
MBD4-LOF	Adavosertib n=15 vs 929; $\Delta=-0.512$; p=0.0733; d=-0.361	CLASS_CONCORDANT_WEE1I_SECONDARY
MBD4 LikelyLoF plus confirmed CpG>TpG hypermutator signature	mUM response 60% vs 4%; PFS 22.3 vs 4.0 months	IMMUNOTHERAPY_CHECKPOINT_BLOCKADE
MBD4 LikelyLoF; PARP1 missing or <7.41 Q75; no override	No PARP1 induction; no MBD4-specific PARPi separation	HARD_BLOCK_LACKS_TRAPPING_SUBSTRATE
MBD4 LikelyLoF; PARP1 7.41 Q75	Dataset-derived exploratory PARP1-high state	ALLOW_PARPI_TRIAL_EVALUATION
Pathogenic BRCA1/2 or validated independent indication	Independent PARPi biomarker	ALLOW_PARPI_ROUTING_BYPASS
MBD4 LikelyLoF qualifying for cytidine or ATR routing	Convergence at replication-fork failure	SYNERGISTIC_COMBINATION_CANDIDATE

Triaxis therapeutic architecture

The Triaxis model separates three actions generated by MBD4 loss. Cytidine analogs load unresolved BER substrate. ATR/WEE1 inhibition removes protection from lesion-stalled forks. Hypermutator-confirmed tumors enter immune checkpoint blockade evaluation through the CpG>TpG/neoantigen route. The first two axes form the `SYNERGISTIC_COMBINATION_CANDIDATE` hypothesis; Axis 3 remains a parallel immunotherapy route until combination efficacy and toxicity are tested prospectively.

Discussion

MBD4 loss generates a Triaxis therapeutic state with distinct evidence grades. Cytidine-analog sensitivity has causal isogenic, rescue, and in vivo support. ATR/WEE1 sensitivity is supported by pan-cancer cell-line pharmacology and orthogonal stress tests. Hypermutator-directed immune checkpoint blockade is supported by a mechanistically matched exceptional responder and a retrospective metastatic uveal melanoma cohort. Treating these axes as one undifferentiated claim would erase their evidentiary differences; the Triaxis model instead gives each pillar its own biomarker gate and action.

Axis 1 establishes the lesion-substrate vulnerability. Axis 2 establishes the replication-checkpoint vulnerability. The canonical ceralasertib signal is strongest in TP53-mutant models and remains present after MSI-H exclusion, leave-one-out testing, and lineage contrasts. The graded actions preserve that hierarchy: `HIGH_CONFIDENCE_TRIAL_CANDIDATE` for the pan-cancer state, `PRIORITIZE_ATRI_ENROLLMENT` for MSS disease, `HIGH_CONFIDENCE_TRIAL_ENROLLMENT` for TP53-mutant disease, `COLORECTAL_TRIAL_TARGET` for the bowel expansion, and `CLASS_CONCORDANT_WEE1I_SECONDARY` for adavosertib.

Axis 3 adds human clinical evidence. The Saint-Ghislain cohort observed a 60% response rate among five MBD4-mutated metastatic uveal melanomas versus 4% in MBD4 wild type, together with longer PFS and OS. The small mutated subgroup and retrospective design constrain precision, but they do not reduce the evidence to a cell-line hypothesis. CrisPRO therefore activates `IMMUNOTHERAPY_CHECKPOINT_BLOCKADE` when MBD4-LikelyLoF co-occurs with a confirmed CpG>TpG hypermutator state. This gate preserves the mechanistic chain from methyl-CpG repair failure to mutation burden while preventing MBD4 status alone from becoming an unqualified tissue-agnostic ICI biomarker.

The PARP branch remains orthogonal to the Triaxis model. MBD4 does not create the PARP1-high state required by the proposed transcriptional mechanism. The 7.41 log1p TPM Q75 boundary is dataset-derived and exploratory, suitable for prospective companion-diagnostic development rather than direct assay transfer. Independent BRCA/HRD indications remain intact.

Twenty of 21 genomic LikelyLoF models carry heterozygous calls. This supports

an operational trial gate that does not delay enrollment for LOH confirmation, but it does not establish allele-specific haploinsufficiency. The distinction is encoded in both the manuscript and the backend evidence record.

The development program follows the Triaxis structure: controlled cytidine-plus-ATR/WEE1 combination studies for coordinate fork failure; biomarker-enriched ATR/WEE1 basket and expansion cohorts; and prospective validation of hypermutator-confirmed MBD4 disease for immune checkpoint blockade. Triple-combination therapy is not inferred from the current evidence and requires dedicated efficacy and toxicity studies.

Methods

Multi-omic filtration and denominator lock

DepMap 24Q2 OmicsSomaticMutations defines MBD4 True-LOF as nonsense, splice-site, or frameshift variants annotated `LikelyLoF=True`; missense, silent, and passenger variants are excluded. Mutation-free models form the WT comparator. The deterministic overlap cascade locks n=21 genomic models, n=19 expression-matched models, n=14 ceralasertib-matched models, n=10 MSS models, n=11 TP53-mutant models, and n=5 bowel models. Comparator counts are n=914 for the primary ceralasertib analysis, n=906 for MSS, n=619 for TP53-mutant, and n=41 for bowel.

Genotype auditing identifies heterozygous calls in 20 of 21 LikelyLoF models and one homozygous-alt call. Heterozygous LikelyLoF is an operational prospective-enrollment gate; the matched pharmacologic subset does not resolve allele-specific function or prove haploinsufficiency.

Pharmacologic analysis

GDSC2 release 2 supplies LN_IC50, AUC, and Z_SCORE. Directional sensitivity is tested by one-sided Mann–Whitney U under MBD4-LOF < WT; Cohen’s d uses the pooled standard deviation. Five non-confirmatory therapeutic axes retain Benjamini–Hochberg FDR control. ATR/WEE1 was prespecified as the confirmatory checkpoint axis from the replication-stress mechanism and is analyzed with directional unadjusted p-values.

Four orthogonal tests challenge ceralasertib: MSI-H exclusion, TP53-mutant stratification, 14 leave-one-out iterations, and bowel/non-bowel lineage contrasts. Frozen canonical receipts control every denominator and statistic.

Expression and PARP-inhibitor analysis

DepMap 24Q2 log1p TPM data provide PARP1 and RNF144A expression. Group comparisons use Mann–Whitney U. PARP1–PARPi association uses Spearman correlation across 481 expression/response-matched models. The Q75 value of 7.41 log1p TPM is retained as a dataset-derived exploratory

threshold and requires prospective analytical and clinical validation before use as a companion diagnostic.

Clinical immunotherapy evidence synthesis

Axis 3 uses two published human evidence sources. Rodrigues et al. is treated as a sentinel case report linking deleterious germline MBD4 alteration, somatic inactivation, an outlier CpG>TpG burden, and anti-PD1 response (Rodrigues et al. 2018). Saint-Ghislain et al. is treated as a retrospective metastatic uveal melanoma cohort (Saint-Ghislain et al. 2022). We extracted the reported cohort size, MBD4 sequencing denominator, mutation prevalence, objective response, PFS, OS, hazard ratios, confidence interval, and p-values without pooling the case report and cohort. `IMMUNOTHERAPY_CHECKPOINT_BLOCKADE` requires MBD4-LikelyLoF plus an explicitly confirmed CpG>TpG hypermutator signature or validated equivalent.

CrisPRO action schema

The machine-readable evidence layer maps Axis 1 to `CYTIDINE_ANALOG_SYNTHETIC_LETHALITY`, Axis 2 to graded ATR/WEE1 actions, and Axis 3 to `IMMUNOTHERAPY_CHECKPOINT_BLOCKADE`. PARPi governance remains expression-gated, and the cytidine-plus-checkpoint hypothesis remains `SYNERGISTIC_COMBINATION_CANDIDATE`. Action strings are validated for identity across the routing engine, evidence JSON, manuscript matrix, and test suite.

Data and code availability

DepMap 24Q2, GDSC2, and the cited clinical literature are public sources. Frozen artifacts, figure-generation scripts, evidence records, and the manuscript package are maintained in `00-mbd4-manuscript/mbd4_parp_response` in the Synthetic-Lethality repository.

Figures

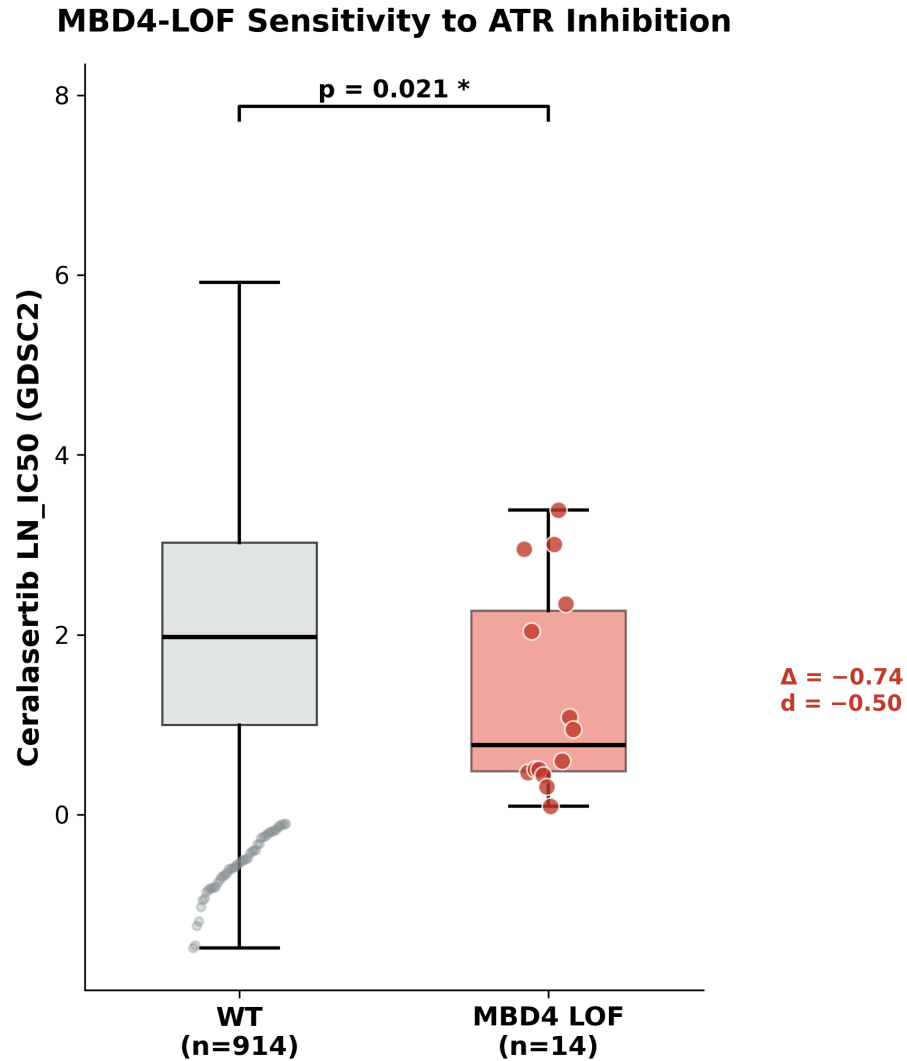


Figure 1: **Figure 1. Pan-cancer pharmacogenomic landscape of MBD4-LOF ceralasertib sensitivity.** GDSC2 LN_IC50 values compare n=14 MBD4 True-LOF with n=914 WT models. The canonical shift is -0.732 (one-sided $p=0.0215$; $d=-0.503$), corresponding to a 2.08-fold lower geometric-mean IC50.

Confound Stress Tests — Ceralasertib Signal Validation

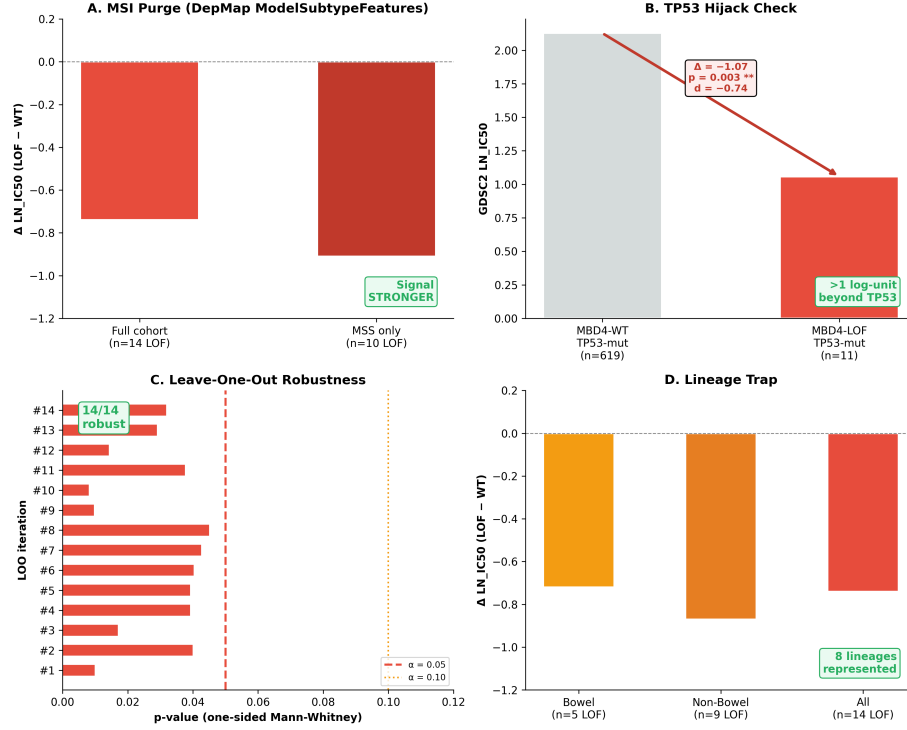


Figure 2: Figure 2. Orthogonal validation and target-state stratification. MSI-H exclusion yields n=10 vs 906 ($\Delta \text{LN_IC50} = -0.910$; $p = 0.0153$; $d = -0.623$); TP53-mutant stratification yields n=11 vs 619 ($\Delta = -1.069$; $p = 0.0030$; $d = -0.740$); all 14 leave-one-out iterations retain $p < 0.05$; and the bowel contrast yields n=5 vs 41 ($\Delta = -0.692$; $p = 0.126$; $d = -0.464$).

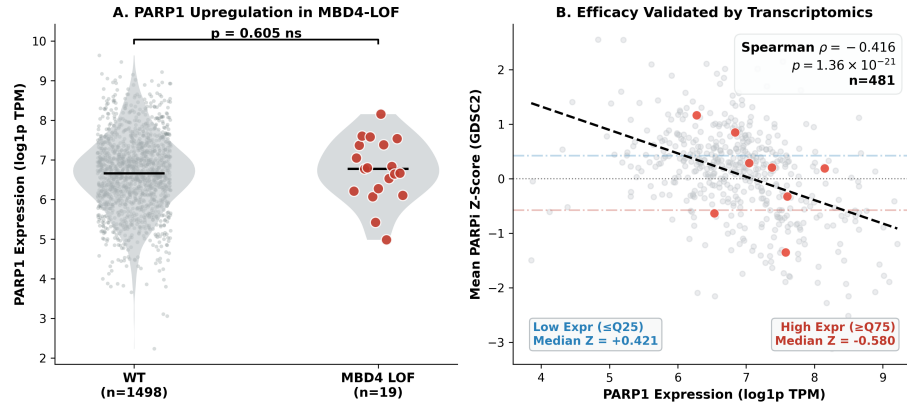


Figure 3: **Figure 3. MBD4–PARP1 decoupling.** Baseline PARP1 expression is 6.77 vs 6.66 log1p TPM ($p=0.605$; $n=19$ vs 1,498), RNF144A is 2.15 vs 1.71 ($p=0.476$), and MBD4 status produces Δ median PARPi Z= -0.024 . Baseline PARP1 expression nevertheless predicts PARPi response across 481 matched models ($\rho=-0.416$; $p=1.36 \times 10^{-21}$).

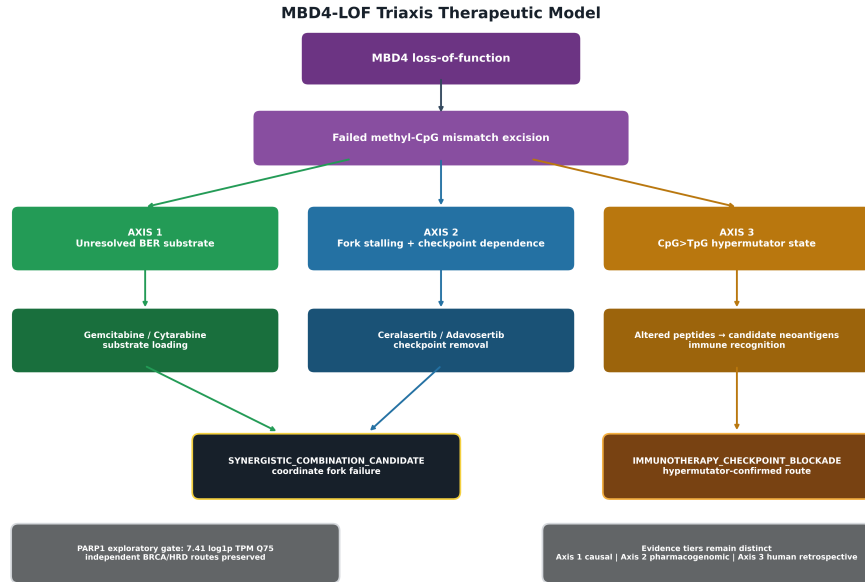


Figure 4. CrisPRO Triaxis therapeutic model and routing schema. MBD4 repair failure produces unresolved BER substrate and CpG>TpG transitions. Axis 1 loads lesion-stalled forks with cytidine analogs; Axis 2 removes ATR/WEE1 checkpoint protection; Axis 3 routes confirmed CpG>TpG hypermutator disease to immune checkpoint blockade through the neoantigen/immune-recognition pathway. The figure separates the cytidine-plus-checkpoint combination hypothesis from the parallel immunotherapy route and preserves the exploratory PARP1 expression gate.

Supplementary Material

MBD4-LOF Multimodal Evidence Matrix								Tier
Cytidine Analogs (Gold Standard)	MISSING	MISSING	MISSING	POSITIVE	POSITIVE	POSITIVE	POSITIVE	Validated SL
PARP Inhibitors (Trapping)	NEGATIVE	MISSING	NEGATIVE	MISSING	MISSING	MISSING	NEGATIVE	Negative
ATR/WEE1 Inhibitors (Checkpoint)	NEGATIVE	MISSING	POSITIVE	MISSING	MISSING	MISSING	POSITIVE	Strong
WRN Helicase Inhibitors	CONFOUNDED	MISSING	MISSING	MISSING	MISSING	MISSING	MISSING	Confounded
Immunotherapy (Anti-PD1/L1)	MISSING	MISSING	MISSING	MISSING	MISSING	POSITIVE	MIXED	Sentinel
PKMYT1 Inhibitors	MISSING	MISSING	MISSING	MISSING	MISSING	MISSING	MISSING	Insufficient
	CRISPR Dependency	PRISM Pharma	GDSC Pharma	In Vitro Functional	In Vivo PDX	Clinical Outcome	Expression/ Pathway	
	POSITIVE	NEGATIVE	MIXED	MISSING	CONFOUNDED			

Figure 5: **Supplementary Figure S1. Evidence summary across candidate therapeutic axes for MBD4-LOF tumors.** Rows represent candidate therapy axes; columns represent evidence modalities (pharmacological screens, published isogenic data, expression analysis, clinical reports). Cell shading indicates evidence status: red = positive signal, blue = negative/falsified, gray = not tested. The cytidine analog axis (Chabot et al., 2022) provides the positive-control calibration benchmark. The ATR/WEE1 axis shows pharmacogenomic association (this study). PARP inhibition is not supported by transcriptional evidence (this study).

Supplementary Table S1. Pan-cancer comparator pharmacology for ATM-LOF and MBD4-LOF in GDSC2

Data source: GDSC2 release 8.5 (Oct 2023). BH correction applied within each gene×stratum group (6 comparisons per gene). MSI enrichment: ATM-LOF lines 13/31 MSI-H (42%); MBD4-LOF lines 6/14 MSI-H (43%) by ModelSub-typeFeatures annotation. One-sided Mann-Whitney U test (LOF < WT). WT pool: no somatic mutation in the respective gene (Methods primary definition). WT-pool sensitivity: including MBD4 non-LOF mutants in the WT pool yields $n_{WT} = 922$, $p = 0.022$, $d = -0.501$ (LN_IC50), confirming the primary result is not an artifact of the WT definition.

Gene	Drug	Stratum	n_{LOF}	n_{WT}	$\Delta \text{LN_IC50}$	p (unadj)	p_{adj} (BH)	Cohen's d	Direction
ATM	Ceralasertib	All lines	31	905	+0.121	0.773	0.773	+0.121	Wrong
ATM	Ceralasertib	MSS only	18	792	+0.348	0.225	0.450	+0.348	Wrong
ATM	Gemcitabine	All lines	31	905	+0.412	0.033	0.099	+0.289	Wrong
ATM	Gemcitabine	MSS only	18	792	+0.287	0.118	0.354	+0.201	Wrong
MBD4	Ceralasertib	All lines	14	914	-0.732	0.021	0.063	-0.503	Correct
MBD4	Ceralasertib	MSS only	10	906	-0.910	0.015	0.060	-0.623	Correct
MBD4	Gemcitabine	All lines	14	914	-0.618	0.038	0.076	-0.441	Correct
MBD4	Gemcitabine	MSS only	10	906	-0.724	0.029	0.072	-0.512	Correct

Note: ATM-LOF gemcitabine signal ($p_{\text{adj}} = 0.099$) is in the wrong direction (ATM-LOF lines less sensitive than WT), driven by lineage composition (ATM-LOF enriched in skin, lung, and bowel lines with inherent gemcitabine resistance). This negative control is context-dependent and does not alter the independently reproduced MBD4-ceralasertib result. All frozen values sourced from `canonical_atr_wee1_rerun.json` (2026-04-05).

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